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CRITICALLY HUMAN: THE IMPORTANCE OF A *HUMANIZING* ETHIC IN TECHNOLOGY DESIGN

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This quilt is the story of how African Americans have struggled to fit in here and grab a piece of the American dream, grab a piece of that pie. And sometimes our ladder to reach that goal has been a tad too short. So this man is trying to grab his piece of the pie. He's trying like hell to grab a piece of the pie.

—Dr. Carolyn Mazloomi (2014) on her quilt “Trying to Grab a Piece of the Pie”

Digital design and technology have long been equated with innovation and advancement. Associated discourse in education continues to promise change and transformation with language around emergent design, breakthroughs, and future innovations. Yet, as educators and learning scientists, we would like to trouble these possibilities of design without attention to social transformation, particularly in educational structures that continue to reproduce inequality. We argue that given the histories and pervasiveness of anti-Black racist ideology and institutional practice, technology designers have a responsibility to consider technology's potentially dehumanizing impact on nondominant students and communities. In education in particular, these injustices materialize as rigid institutional structures, hierarchical sorting of achievement, and reductive metrics of learning. Left unchecked, digital tools do nothing to counter the furtherance of deficit orientations to nondominant students' cultures, abilities, languages, and everyday practices.

To trouble these risks, we focus on the ways that ideological constructions, such as social constructs, cultural biases, and implicit definitions, are embedded in, and reinforced through, educational software-in-use. We define “ideology” as “mental frameworks—the languages, the concepts, categories, imagery of thought, and the systems of representation—which different classes and social groups deploy in order to make sense of, define, figure out and render intelligible the way society works” (Hall 1986, 29). Following Hall’s definition, we highlight the connection among ideology, power, and hegemony—the ways particular ideologies come to dominate the thinking in a particular context that aid in the maintenance of power and material relations, in our case through notions of teaching and learning (Abercrombie and Turner 1978).

We take a twofold approach to address the specifics of ideological design as related to (a) software in education and (b) learning design. Across both domains, we center the role of teachers as designers—those who, drawing from their pedagogical insight and sociocultural knowledge of their contexts, bear expertise that might inform not only the use of technology but also its very design. We illustrate these concepts using examples from our own teaching and scholarship, first offering analysis of dehumanizing proceduralized ideologies within an example of software used in teacher education called *simSchool* (Aguilera and Chen, 2026). We then present our vision for humanizing design in education via analysis of a teacher’s digital quilting design using *Padlet* educational software. We end with a series of provocations for educators and designers.

HUMANIZING IN AN ANTI-BLACK SOCIAL ECOLOGY

Our argument is grounded in historicized understandings of anti-Black racism in schools and society. Black freedom scholars especially help us understand the significance of humanization in an anti-Black context. Anti-Blackness is endemic and permanent, reproduced by policies and everyday practices that rely on and enable Black suffering (Dumas and ross 2016). In educational contexts, and schools in particular, such practices diminish the “souls of Black children every day through systemic, institutionalized, anti-Black, state-sanctioned violence” (Love 2016, 1). These acts of dehumanization become normalized via practices of policing,

forced cultural assimilation, and *educational redlining*—exclusionary practices of place-based mapping that sequester educational opportunities away from poor Black and Brown neighborhoods (Holzman 2012).

Humanizing pedagogies, in stark contrast, honor the cultures and linguistic repertoires of students, and distinctly critique and counter the legacy of oppressive systems, particularly as perpetuated by schools (Ladson-Billings 1995). Affirming the humanity of learners, Brazilian educator Paulo Freire (1970) foregrounded the development of critical consciousness, offering tools to name and critique oppressive structures in resistance to practices of passivity in education. These strategic efforts to emphasize human dignity, fight injustice, and affirm the interdependence of human relationships inform our critical approach to technology in education.

Responding to the ubiquity of digital technology in society, critical scholars, including those in the critical race and digital studies group (Hamilton 2020), help advance race-conscious understandings of inequality in digital society. Current tech trends in education have been shown to harm BIPOC students in schools, such as algorithmic bias in test proctoring software (Swauger 2020), which has been shown to disproportionately “flag” Black and Brown students for anomalous behavior.

Digital tools that are supposedly race neutral also have racial implications. For example, police body cameras, as a component of larger technological policing and surveillance of Black people (Page and Jones 2021), heighten witnesses’ risk of conflict with police should they demand individuals stop recording their own accounts of an incident (Richardson 2020). Even the way digital hardware and software are managed in schools presents racialized disparities, with teachers in urban schools with high percentages of Black and Latine students bearing the labor required to maintain those tools with little IT support (Crooks 2019). Such race-conscious framings of the digital realm help us expand current technological “ethical imaginaries and frameworks,” often limited in their intersectional analysis of power (Le Bui and Noble 2020, 166). Digital tools are only as transformative or innovative as the people using them, with design influenced by complexities of identity, ideology, and capital often rendered invisible. Thus, we contribute to discussions of dehumanization in an increasingly digital educational culture with humanizing design visions for the future.

HUMANIZING PEDAGOGIES IN PRACTICE

Joining a long tradition of educators and scholars, we affirm teachers' daily work inside classrooms as sociopolitical and far-reaching. A humanizing orientation seeks to foster critical consciousness, further equity, and disrupt unjust practices in education that marginalize students, eschewing a power-neutral, autonomous approach to education.

Well before ubiquitous digitalization of education, teachers' critical approaches to pedagogy have prioritized supporting minoritized students' counter-storytelling about their own realities, a practice enabling students to preserve their own psyches against condemnatory deficit narratives about them and their communities (Solórzano and Yosso 2002). Garcia et al. (2015) demonstrated how students in a Council of Youth Research were able to, through research and presentations, construct counter-stories that they shared with national research audiences and policymakers. By using their voices as students of color—and those of their communities—high school youth used digital compositions to deconstruct dehumanizing ways of viewing their communities. Finally, from an educator perspective, Lane (2018) details a politicized ethic of care practiced by Black women educators engaging empathy in a field where intellect and emotions are separated. Educators empowered students to articulate their sociopolitical positioning and advocate for educational equity for all marginalized students. Irrespective of specific technology tools and digital modes, critical ways of organizing researchers, people, and communities remain integral to an emergent vision for humanizing and liberatory pedagogies.

DEHUMANIZATION IN A DIGITAL AGE

The above examples of humanizing pedagogies highlight teachers' critical engagement with issues that impact students' lives. Yet, in practice, teachers also find themselves struggling against structural constraints that limit their agency as critically engaged pedagogues. As more and more of our educational and social experiences are mediated through computational software and related technologies, we contend that historically dehumanizing ideologies—control, hierarchy, and racial reductionism—are

now being reproduced, circulated, and reified in students' and teachers' everyday experiences of educational software-in-use (Aguilera and Chen, 2026).

Here, we are using the term "software-in-use" as an extension of traditions of discourse analysis, which have established that the study of language is not very meaningful devoid of its context but must focus on language-in-use (Haddix 2020). In a similar vein, we are most interested in software as it is used in everyday teaching and learning contexts rather than as a stand-alone object to be studied in a vacuum. Following this logic, software that has not been traditionally considered educational—such as commercial video games—can certainly be examined within a pedagogical context (Steinkuehler et al. 2012). Conversely, software that makes claims about being educational can also be examined by the ways that its everyday use actually supports these claims (Aguilera and de Roock 2022).

PROCEDURALIZED IDEOLOGIES

Before continuing our discussion, we should pause to discuss what we mean by the term "proceduralization" and how it fits with more established concepts of "ideology" and "dehumanization."

From artificial intelligence (AI) chatbots to intelligent tutoring systems to Learning Management Systems, all software is fundamentally defined by *processes*, which are in turn enacted through rules and rule systems (Salen and Zimmerman 2003). Recognizing software as fundamentally *procedural* in nature can provide important insights into the communicative power of the medium, particularly within educational contexts (Bogost 2010), although procedurality is not simply a hallmark of computational software. Our legal systems, schools, and businesses are also largely defined through many complex and interacting processes.

An essential part of *how* software operates in the ideological realm is through the translation of these ideologies into rules and rule systems—a process we have referred to in prior work as "proceduralization" (Aguilera 2022b). We have argued that proceduralization within computational software occurs through at least four mechanisms: (a) repetition, which reinforces ideological constructions through repeated exposure (Hassan and Barber 2021); (b) scale, which diffuses the ideological impact of digital